

Arya Bhavsar

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EDUCATION

The Ohio State University || Columbus, OH

August 2022 – December 2025

Bachelor of Science in Computer Science and Engineering

Specialization in Artificial intelligence

- GPA – 4.00/4.00, Summa cum laude. Dean's List: Autumn 2022 – Fall 2025
- Completed the STEM Exploration and Engagement Scholars Program

WORK EXPERIENCE

The Ohio State University: CSE Undergraduate Teaching Assistant

August 2023 – December 2024

- Delivered instruction on core programming principles in Java for Software 2, a data structures and algorithms course, including Object-Oriented Programming (OOP) principles, and Git for professional workflows.
- Provided support to students during lab sessions and office hours by clarifying technical concepts, debugging Java Programs, and assisting with testing and correctness issues in assignments and projects.
- Drove continuous improvement by evaluating homework, labs, and projects with constructive feedback.

PROJECTS

Events Recommendation Chatbot | *React, Python, FastAPI*

- Designed a full stack chatbot application using React and Tailwind CSS for the frontend, and FastAPI for backend services following a RESTful API architecture.
- Designed API endpoints to process user inputs, generate vector embeddings, and return JSON responses for real-time personalized recommendations based on user inputs.
- Integrated ChromaDB vector database, and LLM APIs Cohere and Gemini Flash with the LangChain framework to enable semantic search and deliver personalized recommendations.

CSE Teaching Assistant Management App | *Ruby on Rails, HTML/CSS, SQLite*

- Developed full-stack web application using Ruby on Rails following MVC architecture to manage teaching assistants for OSU's Department of Computer Science and Engineering.
- Designed and maintained relational database schema and backend logic, enabling CRUD operations for user, course, and section management.
- Implemented REST API endpoints with HTTParty gem and parsed JSON responses for courses data. Implemented user authentication and role-based access for admins, instructors, and students using Devise gem.

Fitness Tracking App | *SwiftUI, Firebase, NoSQL*

- Built a native iOS application using SwiftUI with a clean MVVM architecture to manage state and UI updates.
- Designed a NoSQL database using Firebase/Firestore to securely store user data, workout plans, and daily exercise progress using collections and document-based data modeling. Integrated Firebase Authentication for secure user authentication and access control.
- Built state-driven, reactive UI components leveraging SwiftUI data binding to reflect live updates from Firestore.

Machine Learning Projects | *Python, NumPy, PyTorch, Scikit-learn*

- Trained Naive Bayes classifier and used Linear Discriminant Analysis (LDA) to detect spam emails.
- Implemented k-means clustering algorithm for image compression while preserving image quality.
- Implemented a Support Vector Machine with both batch and stochastic gradient descent.
- Used Principal Component Analysis (PCA) on the Yales Faces dataset to extract facial features.

SKILLS

- **Programming Languages:** Java, C, Python, SQL, Ruby, HTML, CSS, JavaScript, Swift
- **Frameworks and Libraries:** React, Tailwind CSS, FastAPI, PyTorch, Scikit-learn, Ruby on Rails, SwiftUI
- **Tools:** JUnit, Git, GitHub, Docker, Docker Compose, Xcode
- **Technologies:** RESTful APIs, SQLite, PostgreSQL, Supabase, NoSQL, Cloud Firestore